TITLE

Development and validation of a transcriptome-based prognostic signature for survival prediction in lung squamous cell carcinoma.

AUTHOR

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ABSTRACT

Background:
Lung squamous cell carcinoma (LUSC) remains associated with poor clinical outcomes and limited prognostic biomarkers. The aim of this study was to develop and validate a transcriptome-based prognostic model for overall survival in LUSC.

Methods:
Transcriptomic and clinical data from TCGA-LUSC were analyzed using a train/test design including 450 patients (315 training and 135 test samples). An elastic-net Cox model was first developed, followed by a Cox-prefilter feature selection strategy. Model performance was assessed using Kaplan–Meier analysis, hazard ratios, concordance index, time-dependent ROC analysis, multivariable Cox regression, calibration analysis, decision curve analysis, and external validation using GEO datasets.

Results:
The original elastic-net model achieved concordance indices of 0.656 in the training cohort and 0.578 in the test cohort. The hazard ratio for high-risk versus low-risk patients was 2.234 in the training cohort ($p = 4.49 \times 10^{-6}$) and 1.448 in the test cohort ($p = 0.174$). The Cox-prefilter model selected 50 genes and showed improved prognostic performance with a concordance index of 0.648 and a hazard ratio of 2.428 (95% CI 1.811–3.257, $p = 3.14 \times 10^{-9}$). Multivariable Cox regression demonstrated that the molecular risk score remained independently associated with survival External validation demonstrated significant survival stratification in GSE30219 (n = 289, C-index = 0.604, $p = 2.82 \times 10^{-6}$) and a weaker trend in GSE37745 (n = 196, C-index = 0.498, $p = 0.079$).

Conclusions:
The Cox-prefilter transcriptomic signature demonstrated improved prognostic discrimination, independent prognostic value, and partial reproducibility in external cohorts. These findings suggest that the proposed model may provide a useful framework for survival risk stratification in lung squamous cell carcinoma.

KEYWORDS

Lung squamous cell carcinoma; LUSC; prognosis; gene expression; Cox regression; survival model; external validation

INTRODUCTION

Lung cancer remains the leading cause of cancer-related mortality worldwide, and lung squamous cell carcinoma (LUSC) represents a major histological subtype characterized by substantial biological and clinical heterogeneity. Despite advances in systemic therapy and molecular profiling, prognostic stratification in LUSC still relies primarily on conventional clinicopathological variables such as tumor stage, age, and performance status. These variables provide only limited resolution in predicting individual patient outcomes.

Recent developments in high-throughput transcriptomic technologies have enabled the systematic identification of molecular signatures associated with tumor behavior and patient survival. Integrative bioinformatics approaches using publicly available datasets, such as The Cancer Genome Atlas (TCGA), provide valuable opportunities to discover and validate prognostic biomarkers.

In this study, we developed and evaluated transcriptome-based survival prediction models for patients with LUSC. An elastic-net Cox regression framework was first applied within a train/test design to construct a prognostic model. To improve biological interpretability and robustness, we further implemented a Cox-prefilter feature selection strategy to identify survival-associated genes. The prognostic performance of the resulting models was evaluated using Kaplan–Meier survival analysis, concordance index estimation, and time-dependent ROC analysis. We additionally assessed the independent prognostic value of the molecular risk score using multivariable Cox regression including clinical covariates. Finally, the proposed signature was externally validated in independent Gene Expression Omnibus (GEO) cohorts, and its potential clinical relevance was explored through calibration analysis and decision curve analysis.

MATERIALS AND METHODS

Data sources and study design

Transcriptomic and clinical data for lung squamous cell carcinoma (LUSC) were obtained from The Cancer Genome Atlas (TCGA) database. RNA-sequencing expression data and corresponding clinical annotations were downloaded using publicly available resources. After quality control and matching of clinical information with expression profiles, a total of 450 patients were included in the analysis.

The dataset was randomly divided into a training cohort (n = 315) and an independent test cohort (n = 135). The training cohort was used for model development and feature selection, while the test cohort was used for independent internal validation of the prognostic model.

Gene expression preprocessing

Gene expression data were processed and normalized prior to analysis. Low-expression genes were filtered to reduce noise and improve model stability. Expression values were log-transformed where appropriate to ensure comparability across samples. Only genes with sufficient expression variability across samples were retained for downstream analyses.

Construction of the elastic-net Cox model

A penalized Cox proportional hazards regression model using elastic-net regularization was implemented to identify prognostic genes associated with overall survival. Elastic-net combines both LASSO and ridge penalties, allowing simultaneous variable selection and coefficient shrinkage.

The model was trained using the training cohort, and optimal penalty parameters were determined through cross-validation. A molecular risk score for each patient was calculated as a weighted linear combination of selected gene expression values multiplied by their respective regression coefficients.

Patients were stratified into high-risk and low-risk groups based on the median risk score.

Cox-prefilter feature selection

To improve biological interpretability and model robustness, an additional Cox-prefilter strategy was applied. Genes significantly associated with survival in univariate Cox regression analysis were first identified. These survival-associated genes were then used as candidate features for the final prognostic model.

This approach reduced the feature space and allowed the identification of a transcriptomic signature consisting of 50 genes.

Survival analysis

Kaplan–Meier survival analysis was used to compare overall survival between high-risk and low-risk patient groups. Statistical significance between survival curves was evaluated using the log-rank test.

Hazard ratios (HR) and 95% confidence intervals (CI) were calculated using Cox proportional hazards regression models.

Multivariable Cox regression

To evaluate the independent prognostic value of the molecular risk score, multivariable Cox regression analysis was performed including clinical variables such as age, gender, and tumor stage. Hazard ratios, confidence intervals, and p-values were reported to assess the contribution of each variable to overall survival.

Model performance evaluation

Model discrimination was assessed using the concordance index (C-index). Time-dependent receiver operating characteristic (ROC) curves were calculated to evaluate the predictive accuracy of the model at different time points (1, 3, and 5 years). Area under the curve (AUC) values were computed for both training and test cohorts.

Calibration analysis

Calibration analysis was performed to evaluate the agreement between predicted risk scores and observed survival outcomes. Patients were grouped according to predicted risk levels, and observed event rates were compared with model predictions.

Decision curve analysis

Decision curve analysis (DCA) was conducted to assess the potential clinical utility of the prognostic model. Net benefit was calculated across a range of threshold probabilities and compared with default clinical strategies, including treat-all and treat-none approaches.

External validation

External validation of the prognostic signature was performed using independent datasets from the Gene Expression Omnibus (GEO). Two cohorts were analyzed:

- GSE30219 (n = 289)
- GSE37745 (n = 196)

Gene expression data from these datasets were processed to match the genes included in the prognostic signature. Risk scores were calculated using the same coefficients derived from the TCGA model, and patients were stratified into risk groups. Kaplan–Meier survival analysis and concordance index estimation were used to evaluate model performance.

Statistical analysis

All statistical analyses were performed using the R statistical environment. Survival analyses were conducted using the *survival* and *survminer* packages. Penalized Cox regression models were implemented using the *glmnet* package. Time-dependent ROC

analysis was performed using the *timeROC* package. Decision curve analysis was performed using the *rmda* package.

A two-sided p-value < 0.05 was considered statistically significant.

Ethics statement

The study utilized publicly available datasets from The Cancer Genome Atlas (TCGA) and the Gene Expression Omnibus (GEO). All data were previously anonymized and publicly accessible. Therefore, no additional ethical approval was required for this study.

RESULTS

Model performance in the TCGA-LUSC cohort

The original elastic-net model demonstrated moderate prognostic discrimination in the TCGA-LUSC cohort. The concordance index was 0.656 in the training set and 0.578 in the test set. Kaplan–Meier survival analysis indicated separation between high-risk and low-risk patient groups.

The hazard ratio for high-risk versus low-risk patients was 2.234 in the training cohort ($p = 4.49 \times 10^{-6}$) and 1.448 in the test cohort ($p = 0.174$).

Performance of the Cox-prefilter model

To improve model robustness, a Cox-prefilter feature selection strategy was applied. The resulting model included 50 genes and demonstrated improved prognostic stratification across the TCGA dataset. The concordance index of the Cox-prefilter model was 0.648 and the hazard ratio for high-risk versus low-risk patients was 2.428 (95% CI 1.811–3.257, $p = 3.14 \times 10^{-9}$).

Multivariable Cox regression analysis

To evaluate whether the molecular risk score provided prognostic information beyond standard clinical variables, a multivariable Cox regression model was constructed including age, gender, and tumor stage. The molecular risk score remained significantly associated with overall survival after adjustment for clinical covariates (Figure 1), indicating that the transcriptomic signature provides independent prognostic information.

External validation in GEO cohorts

The prognostic model was further evaluated in independent GEO datasets. In the GSE30219 cohort (n = 289), the model demonstrated significant survival stratification with a concordance index of 0.604 ($p = 2.82 \times 10^{-6}$). In contrast, validation in the

GSE37745 cohort (n = 196) showed a weaker but directionally consistent survival trend with a concordance index of 0.498 (p = 0.079).

Calibration and clinical utility of the prognostic model

Calibration analysis indicated that predicted risk scores corresponded to observed survival outcomes across ordered patient groups (Figure 2). Decision curve analysis further demonstrated that the proposed model provided a positive net clinical benefit across a range of threshold probabilities compared with treat-all and treat-none strategies (Figure 3).

An integrated overview of the main analytical steps and validation results is presented in Figure 4.

DISCUSSION

In the present study, we developed and evaluated transcriptome-based prognostic models for lung squamous cell carcinoma using TCGA transcriptomic and clinical data. Our results demonstrate that a Cox-prefilter feature selection strategy can improve prognostic stratification compared with a conventional elastic-net survival model.

The Cox-prefilter model achieved improved separation between high-risk and low-risk patient groups and demonstrated slightly higher discrimination as reflected by concordance index and hazard ratio estimates. Importantly, the molecular risk score remained significantly associated with overall survival in multivariable Cox regression analysis, suggesting that the signature provides prognostic information beyond standard clinical covariates.

External validation in independent GEO cohorts provided partial support for the reproducibility of the proposed signature. In the GSE30219 dataset, the model showed significant survival stratification, whereas validation in GSE37745 demonstrated a weaker but directionally consistent trend.

Several limitations should be acknowledged. First, the analyses were based on retrospective public datasets. Second, external validation was constrained by incomplete gene overlap and heterogeneous phenotype annotation across GEO cohorts. Third, functional enrichment analysis did not reveal strong pathway convergence among the selected genes, which may reflect the heterogeneous biology of LUSC.

Despite these limitations, the present study provides a reproducible bioinformatics framework for developing transcriptomic prognostic signatures and highlights the potential value of Cox-based feature selection strategies for survival modeling. Future studies should aim to validate the proposed signature in larger independent cohorts with

standardized clinical annotation and to explore the biological mechanisms underlying the identified prognostic genes.

CONCLUSIONS

The Cox-prefilter transcriptomic signature demonstrated improved prognostic discrimination in TCGA-LUSC, retained significance in multivariable analysis, and showed partial reproducibility in external GEO datasets. The proposed model may therefore serve as a useful framework for molecular risk stratification in lung squamous cell carcinoma, although further validation in prospective cohorts is warranted.

FIGURE LEGENDS

Figure 1. Multivariable Cox regression analysis of the molecular risk score and clinical covariates in the TCGA-LUSC cohort.

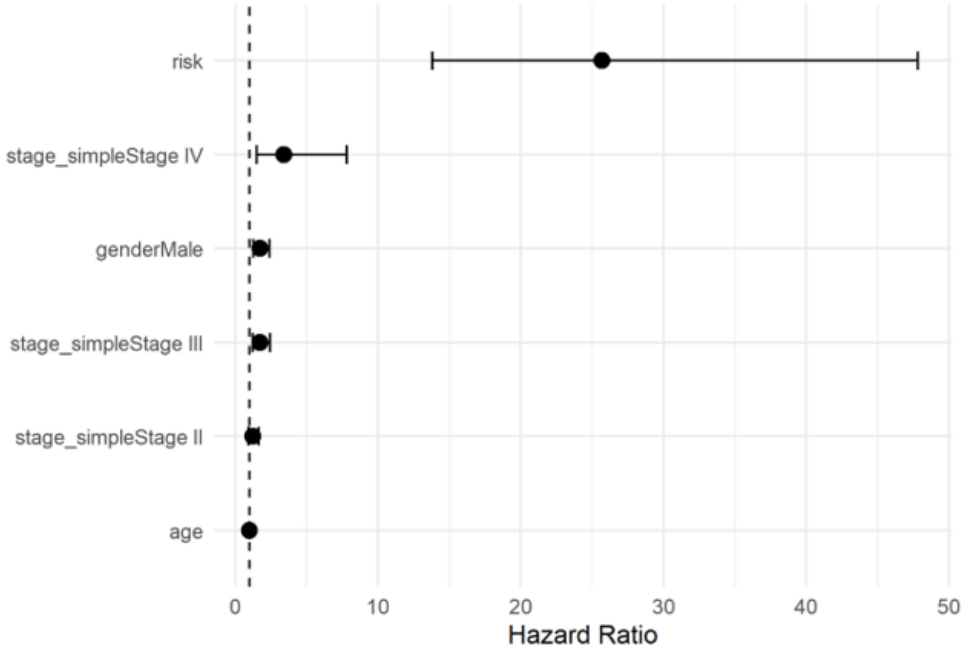

Figure 2. Calibration plot showing agreement between predicted risk scores and observed survival outcomes.

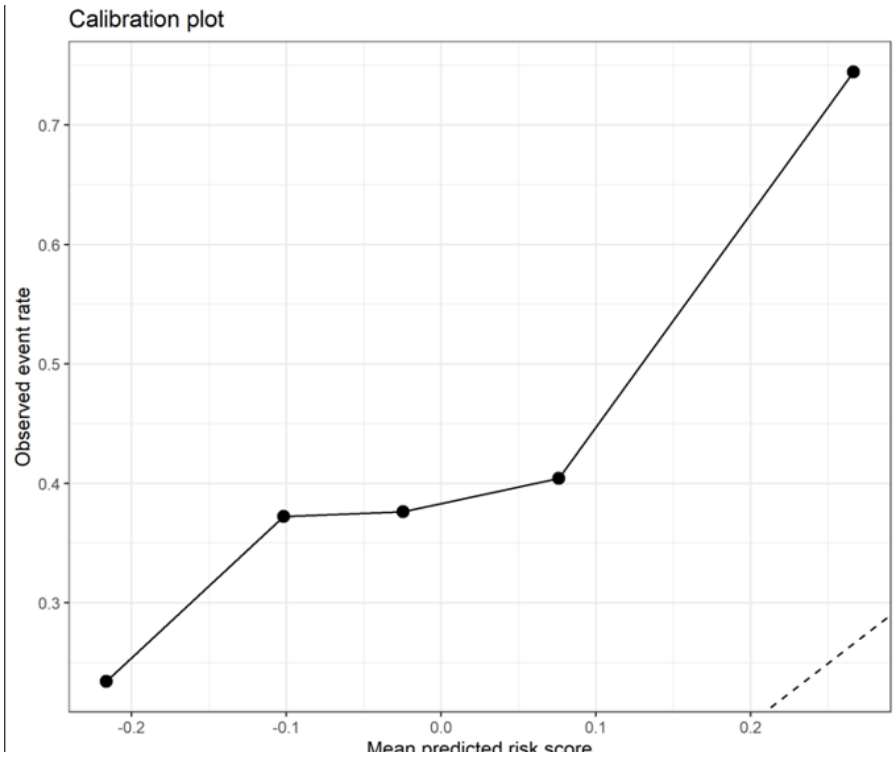

Figure 3. Decision curve analysis evaluating the clinical utility of the prognostic mode

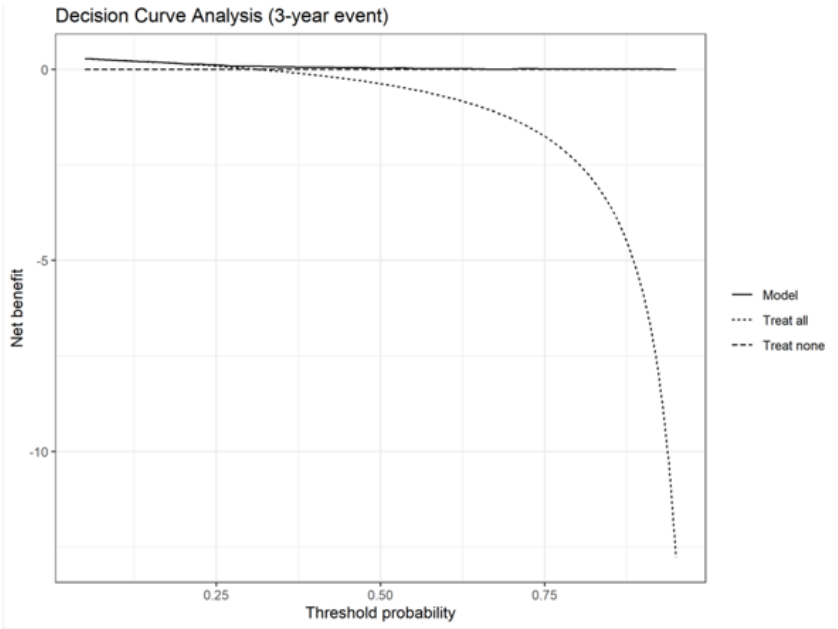
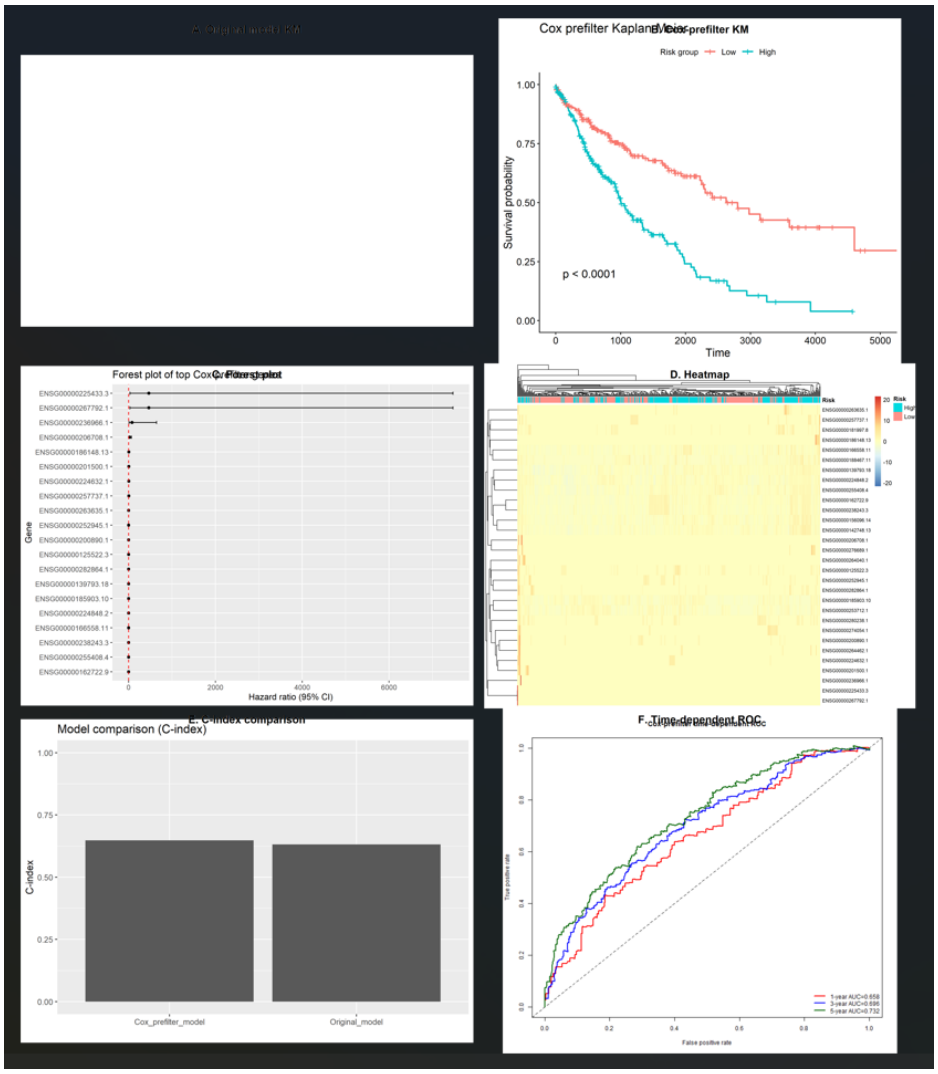
Figure 4. Integrated overview of model performance and validation results.